

Unit I Research methodology

Part-I Introduction -

- **Meaning -** Research refers to a careful, well defined (or re-defined), Objective and systematic method of search for knowledge, or formulation of a hypothesis or p that is driven by Inquisitiveness for that which is unknown and useful on a particular aspect so as to make an original contribution to expand the existing knowledge base.

It comprises -

- defining and redefining problems.
- formulating hypothesis or suggested solutions.
- collecting, organizing, and evaluating data.
- making deductions and reaching conclusions.
- carefully testing the conclusions to determine whether they fit the formulating hypothesis.

• Objectives -

- To solve new and important engineering problems.
- To apply scientific methods to find answers to open questions.
- To develop new theoretical or practical knowledge.
- To explore, describe, diagnose, or test hypothesis.
- To understand reasons for success or failure of an approach.
- To contribute useful insights even if desired results aren't achieved.
- To guide future research and discoveries.

Motivation in engineering research -

- **Intrinsic motivations** - Interest, challenge, learning, meaning, purpose, are linked to creative performance.
- **Extrinsic motivation** - Reward for good work, including money, fame, awards, praise and status.

- Influence from others - competition, collaboration, commitment, and encouragement
- Personal motivation - solving unsolved problems, intellectual joy, service to community, and respectability
- Other related factors - wanting to do better than what has been achieved in world, improve the state of art in technology, contribute to improvement of society, fulfillment of the historical legacy in the immediate sociocultural context.

• Types of engineering research-

→ Descriptive vs. Analytical -

Descriptive - describes the current state or situation (fact finding)

- Uses comparative and correlational methods
- Researchers cannot control variables; only observes and reports
- May try to find causes without controlling factors

Analytical - Uses existing facts or data for analysis and critical evaluation

- Focuses on interpreting and explaining information
- Some studies can be both descriptive and analytical

→ Applied vs. Fundamental -

Applied - Aims to solve immediate, practical problems

- Used in organizations or real life situations
- Example - studying social or economic trends
- Focuses on finding direct solutions for current issues

Fundamental (basic) research -

- Seeks to develop theories or general knowledge
- Involves studying natural phenomena or pure mathematics
- Focuses on long-term understanding and broad applications

→ Quantitative Vs. Qualitative -Quantitative -

- Uses statistical data and large sample sizes.
- Focuses on numerical analysis and measurable results.

Qualitative -

- Uses non-numerical data like verbal descriptions or case studies.
- Based on few, non-representative cases.
- Common in behavioral or observational studies (ex - traffic behavior)

→ Conceptual Vs. Empirical -Conceptual -

- Based on ideas, theories or abstract thinking.
- Focuses on developing new concepts or interpreting existing ones.
- Often used by philosophers and theoreticians.

Empirical research -

- Based on observations, experiments or real-world data.
- Focuses on testing theories or verifying concepts through evidence.
- Common in scientific and practical investigations.

→ Other types of research -

- One-time - done during a single time period.

Longitudinal - carried out over several time periods.

- Field-Setting, Laboratory, or Simulation - based on the environment where research is done.

- Clinical or diagnostic - uses case studies or in-depth methods to find causal relations.

- Conclusion-oriented - flexible; researcher can redesign or modify the study during progress.

Decision-oriented - focused on practical decisions or problem solving.

- **Ethics in research -**

Provides guidelines for the responsible conduct of research.

- Ensure honesty and integrity in conducting and reporting research
- Avoid plagiarism and give proper credit to others' work
- Obtain informed consent from participants before involving them.
- Protect privacy and confidentiality of data and participants
- Avoid falsifying or fabricating data or results
- Follow safety guidelines and minimize harm to people, animals, or the environment.
- Be objective and transparent in analysis and interpretation
- Respect intellectual property and institutional rules
- Ensure social responsibility - research should benefit society

- **Types of research misconduct -**

Research misconduct refers to conducting research in an unethical or dishonest manner. It involves violating standards of integrity, which can harm people, or the environment, waste valuable resources, and undermining the credibility and trustworthiness of research.

- (1) **Fabrication** - making up data or results and recording or reporting them
 - (2) **Falsification** - manipulating research materials, or changing or omitting data or results such that the research is not accurately represented in the research record
 - (3) **Plagiarism** - the appropriation of another person's ideas, processes, results, or words without giving appropriate credit
- Research misconduct does not include honest error or differences of opinion